

Database Design and Implementaion Document

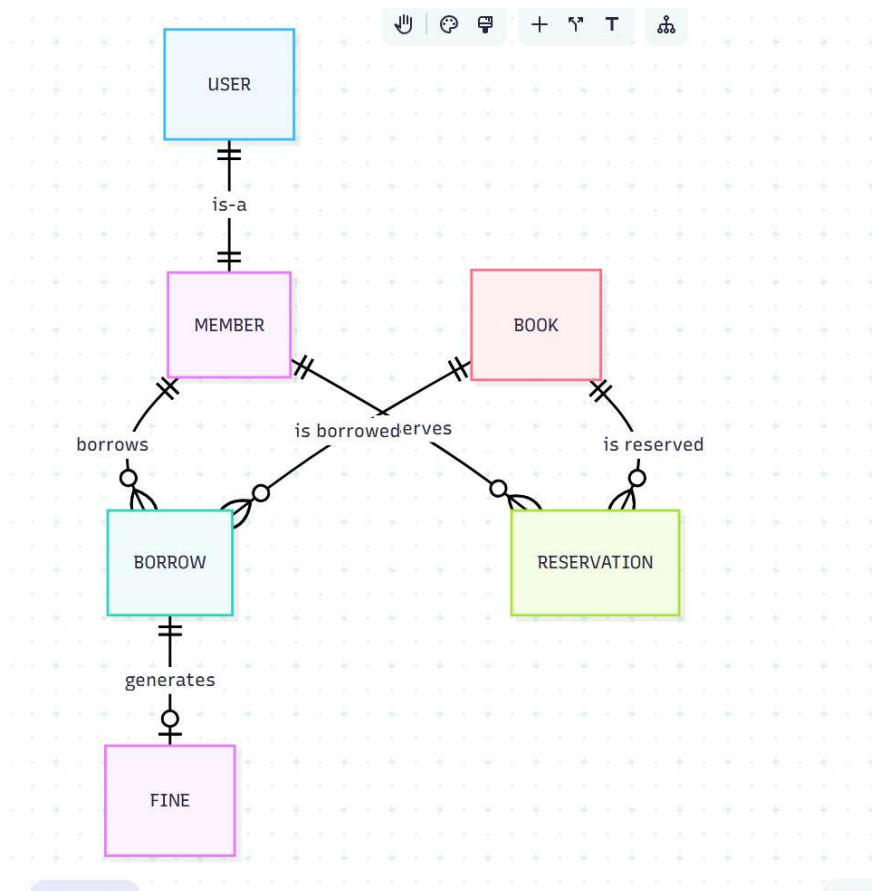
Project Title: Library Management System (LMS)

Project Type: Full-stack Web Development Capstone Project

Document Version: 1.0

Logical Database Diagram

- A member borrows one or more borrowings (a member can borrow multiple items).
- A member reserves zero or more reservations (a member can make multiple reservations).
- Every member is a user, but not every user is necessarily a member
- A book can be borrowed in one or more borrowings (a book can be borrowed multiple times over its lifetime).
- A book can be reserved in zero or more reservations (a book can have multiple reservations).
- A borrowing can generate zero or one fine (a borrowing may result in a fine, but not all borrowings have fines).



Entity-Relationship Diagram

Entity	Left Relation	Right Relationship	Entity	
MEMBER	one-to-one	one-to-many	BORROW	
MEMBER	one-to-one	zero-many	RESERVATION	
USER	one-to-one	zero-to-one	MEMBER	
BOOK	one-to-one	zero-to-many	BORROW	
BOOK	one-to-one	zero-to-many	RESERVATION	
BORROW	one-to-one	Zero-to-one	FINE	

User			
Field	Data type	Key	Contrainst
id	int	PK	
email	string		Unique
password_hash	string		
full_name	string		
phone	phone		/[1-9][0-9]*/
address	string		
role	string		[Admin, Member]

Member			
Field	Data type	Key	Contrainst
user_id	int	PK, FK	
member_id	string		Unique
join_date	date		
status	status		[Active, Suspended, Expired]

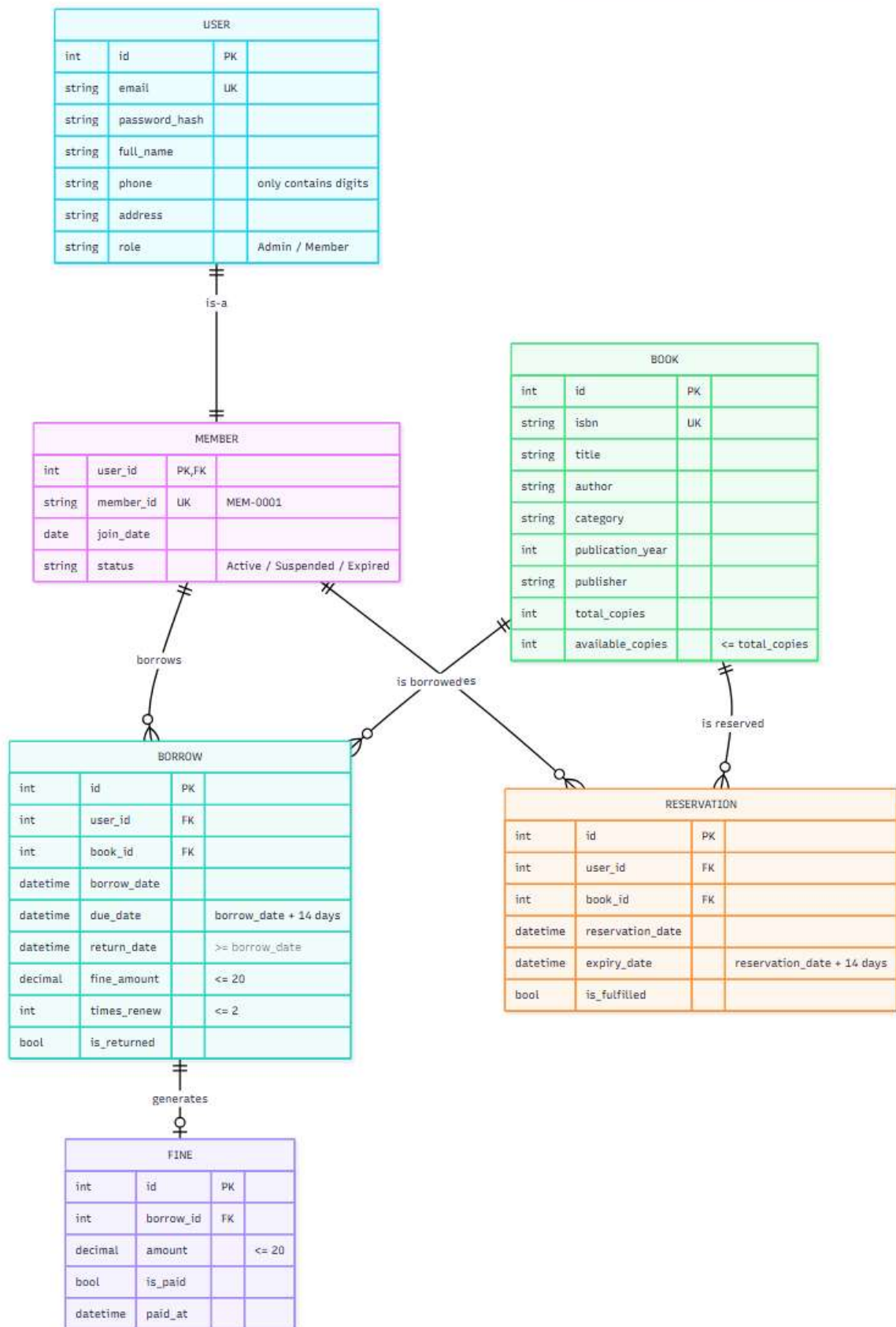
Book			
Field	Data type	Key	Contrainst
id	int	PK	
isbn	string		Unique
title	string		
author	string		

category	string		
publication_year	date (year)		
publisher	string		
total_copies	int		
available_copies	int		<= total_copies

Borrow			
Field	Data type	Key	Constraint
id	int	PK	
user_id	int	FK	
book_id	int	FK	
borrow_date	datetime		
due_date	datetime		borrow_date + 14 days
return_date	datetime		>= borrow_date
times_renewed	int		<= 2 (default 0)
fine_amount	decimal		<= 20.00
is_returned	bool		

Reservation			
Field	Data type	Key	Constraint
id	int	PK	
user_id	int	FK	
book_id	int	FK	
reservation_date	datetime		
expiry_date	datetime		reservation_date + 14 days
is_fulfilled	bool		

Fine			
Field	Data type	Key	Constraint
id	int	PK	
borrow_id	int	FK	
amount	decimal		<= 20.00
is_paid	bool		
paid_at	datetime		



MySQL Statement

```
CREATE TABLE USER (
```

```
    id            INT PRIMARY KEY AUTO_INCREMENT,  
    email         VARCHAR(255) UNIQUE NOT NULL,  
    password_hash VARCHAR(255) NOT NULL,  
    full_name     VARCHAR(255) NOT NULL,  
    phone         VARCHAR(20) NOT NULL,  
    address       TEXT,  
    role          ENUM('Admin', 'Member') DEFAULT 'Member' NOT NULL,
```

```
-- Phone: must contain only digits AND cannot start with 0
```

```
CONSTRAINT CHK_phone_digits_only  
    CHECK (phone REGEXP '^[0-9]+$'),  
CONSTRAINT CHK_phone_no_leading_zero  
    CHECK (phone NOT LIKE '0%')
```

```
);
```

```
CREATE TABLE MEMBER (
```

```
    user_id      INT PRIMARY KEY,  
    member_id    VARCHAR(20) UNIQUE NOT NULL,  
    join_date    DATE DEFAULT (CURRENT_DATE) NOT NULL,  
    status       ENUM('Active', 'Suspended', 'Expired') DEFAULT 'Active' NOT NULL,
```

```
CONSTRAINT FK_MEMBER_USER  
    FOREIGN KEY (user_id) REFERENCES USER(id)  
    ON DELETE CASCADE ON UPDATE CASCADE
```

```
);
```

```
CREATE TABLE BOOK (
```

```
    id            INT PRIMARY KEY AUTO_INCREMENT,  
    isbn          VARCHAR(13) UNIQUE NOT NULL,  
    title         VARCHAR(500) NOT NULL,  
    author        VARCHAR(255) NOT NULL,  
    category      VARCHAR(100),  
    publication_year YEAR NOT NULL,          -- Proper YEAR data type (0000-9999)  
    publisher      VARCHAR(255),  
    total_copies  INT NOT NULL CHECK (total_copies ≥ 0),  
    available_copies INT NOT NULL CHECK (available_copies ≥ 0),
```

```
CONSTRAINT CHK_available_copies  
    CHECK (available_copies ≤ total_copies)
```

```
);
```

```

CREATE TABLE BORROW (
    id INT PRIMARY KEY AUTO_INCREMENT,
    user_id INT NOT NULL,
    book_id INT NOT NULL,
    borrow_date DATETIME DEFAULT CURRENT_TIMESTAMP NOT NULL,
    due_date DATETIME NOT NULL,
    return_date DATETIME NULL,
    fine_amount DECIMAL(6,2) DEFAULT 0.00 CHECK (fine_amount ≤ 20.00),
    fine_paid_at DATETIME NULL,
    times_renew TINYINT DEFAULT 0 CHECK (times_renew ≤ 2),
    is_returned BOOLEAN NULL DEFAULT FALSE,

    CONSTRAINT FK_BORROW_MEMBER
        FOREIGN KEY (user_id) REFERENCES MEMBER(user_id)
        ON DELETE RESTRICT ON UPDATE CASCADE,
    CONSTRAINT FK_BORROW_BOOK
        FOREIGN KEY (book_id) REFERENCES BOOK(id)
        ON DELETE RESTRICT ON UPDATE CASCADE,
    CONSTRAINT CHK_return_date
        CHECK (return_date IS NULL OR return_date ≥ borrow_date),
    CONSTRAINT UQ_USER_BOOK_ACTIVE
        UNIQUE (user_id, book_id, is_returned)
);

```

```

CREATE TABLE RESERVATION (
    id INT PRIMARY KEY AUTO_INCREMENT,
    user_id INT NOT NULL,
    book_id INT NOT NULL,
    reservation_date DATETIME DEFAULT CURRENT_TIMESTAMP NOT NULL,
    expiry_date DATETIME NOT NULL,
    is_fulfilled BOOLEAN DEFAULT FALSE NOT NULL,

    CONSTRAINT FK_RESERVE_MEMBER
        FOREIGN KEY (user_id) REFERENCES MEMBER(user_id)
        ON DELETE CASCADE ON UPDATE CASCADE,
    CONSTRAINT FK_RESERVE_BOOK
        FOREIGN KEY (book_id) REFERENCES BOOK(id)
        ON DELETE CASCADE ON UPDATE CASCADE,
    CONSTRAINT UQ_ACTIVE_RESERVATION
        UNIQUE (user_id, book_id, is_fulfilled)
);

```

```
CREATE TABLE FINE (
    id          INT PRIMARY KEY AUTO_INCREMENT,
    borrow_id   INT NOT NULL,
    amount      DECIMAL(6,2) NOT NULL CHECK (amount ≤ 20.00),
    is_paid     BOOLEAN DEFAULT FALSE NOT NULL,
    paid_at     DATETIME NULL,

    CONSTRAINT FK_FINE_BORROW
        FOREIGN KEY (borrow_id) REFERENCES BORROW(id)
        ON DELETE CASCADE ON UPDATE CASCADE,
    CONSTRAINT CHK_paid_at
        CHECK (is_paid = FALSE OR paid_at IS NOT NULL)
);
```

-- Indexes

```
CREATE INDEX IDX_BOOK_TITLE ON BOOK(title);
CREATE INDEX IDX_BOOK_ISBN  ON BOOK(isbn);
CREATE INDEX IDX_BORROW_DATES ON BORROW(borrow_date, due_date);
```